

Shreya Jain

+1 (734) 450 9984 · shreyadjain26@gmail.com · [linkedin.com/in/shreyadjain/](https://www.linkedin.com/in/shreyadjain/) · github.com/JainShreya26 · shreyadjain.com/

EDUCATION

University of Michigan

Master's in Data Science (Statistics)

Ann Arbor, MI, USA

Aug 2023 - May 2025

WORK EXPERIENCE

Data Scientist

University of Michigan (Michigan Medicine), Ann Arbor, MI, USA

Mar 2025 - Present

- Built domain-specific RAG research app integrating OpenAI, Anthropic, Llama, and MedGemma with Semantic Scholar and Unpaywall pipelines, **increasing citations by 85%**.
- Engineered cross-model ranking system to benchmarking 4 LLMs on **300+** test queries, enabling data-driven model selection.
- Implemented smart model routing to dynamically select lightweight models by query type, **reducing token costs by 60%**.
- Developed LLM-driven causal extraction system converting unstructured text into **NetworkX graphs**, enabling path analysis via natural language queries.
- Automated evaluation of LLM-generated causal graphs against **DECI** and **LiNGAM** baselines using centrality metrics.
- Deployed platform on Streamlit Cloud with structured logging and progress tracking, **cutting debugging time by 40%**.

Data Analyst

University of Michigan, Ann Arbor, MI, USA

Jun 2024 - Dec 2024

- Built **SQL**-based data pipelines to analyze participation trends, driving interventions that **increased response rates by 50%**.
- Built real-time **Power BI** dashboards to track completion metrics, reducing **reporting turnaround by 60%**.
- Designed **reinforcement learning-driven** patient engagement games to capture behavioral data, improving retention and feeding downstream predictive models.

Data Science Intern

Ministry of Defence, DRDO Pune, MH, India

Oct 2022 - July 2023

- Built real-time pose estimation system using **OpenCV** and **MediaPipe** to automate validation of 5 army drill exercises, tested on a custom 7-subject dataset across varying heights, lighting, and distances.
- Engineered **YOLOv8** preprocessing pipeline with auto-cropping, **improving detection accuracy by 98%** and reducing noise.
- Implemented rule-based posture classifier using **joint-angle** and **Euclidean distance** metrics against military standards.
- Deployed a offline **Flask** app for the Government, replacing continuous supervision with real-time corrective feedback.

Data Scientist

Beyond Business Travel, Ireland (Remote)

Aug 2022 - Dec 2022

- Automated multi-format invoice extraction using **OCR** and **NLP**, achieving **98%** accuracy and **saving 6 hours daily**.
- Designed rule-based validation and statistical outlier detection to flag pricing inconsistencies, reducing billing errors by **70%**.

PROJECTS

Soccer Analysis ([Link](#))

Tableau Dashboard, Data Analysis, Data Visualization, Statistics

- Processed 3M+ soccer records across 11 datasets using **SQL** and **Python**, performing ETL, data cleaning, and feature engineering to surface performance trends.
- Built interactive Tableau dashboard with multi-dimensional filtering for player comparison and trend analysis.

Ride-Share ETA Prediction & Anomaly Detection System

Python, R, Feature Engineering, Scikit-learn

- Built and validated **XGBoost** and **LightGBM** models on trip, weather, and traffic data, reducing ETA prediction error by **18%**.
- Implemented **Isolation Forest** models to flag trip anomalies and surface risk factors like route deviations and surge spikes.

Offer and Deal Quality Prediction - Shark Tank India ([Link](#))

Python, R, Machine Learning, Artificial Neural Networks

- Built a 121-pitch Shark Tank India dataset and benchmarked six ML classifiers (**ANN**, **SVM**, **Random Forest**) for offer prediction, with ANN achieving **87.09% F1-score**.
- Designed a 22-rule Mamdani fuzzy logic system in **MATLAB** to categorize deal quality by revenue, valuation, and startup age.

SKILLS

Programming and Data: Python, R, C++, JavaScript, SQL, Pandas, Numpy, Tableau, PowerBI, Altair, NetworkX, Matplotlib, Seaborn

Machine Learning and Deep Learning: PyTorch, Tensorflow, Scikit-learn, Langchain, RAG, Vector DB, Hugging Face, Causal Inference

Collaboration and Cloud: Git, Github, Google Cloud